

Linyi (Tiger) Hou

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EDUCATION

Ph.D. in Aerospace Engineering, University of Illinois Urbana-Champaign
B.S. in AE, Minor in CompE, University of Illinois Urbana-Champaign

Jun 2021 - present
Aug 2017 - May 2021

WORK EXPERIENCE

University of Illinois Urbana-Champaign

Teaching Assistant (AE 202, AE 443, AE 483)

Jun 2021 - present

- ❖ Worked with 2-5 faculty and staff to teach lessons, hold office hours, and improve curriculum for courses in flight mechanics, systems engineering, and UAV controls.
- ❖ Assisted in developing code and instructions for a UAV lab based on C and Python.

Collins Aerospace

Software Engineering Intern

May 2019 - Aug 2019

- ❖ Migrated an avionics data interpreter from 16-bit to 32-bit using C++ and Python.
- ❖ Automated builds in TortoiseSVN using Python, reducing need for manual integration.

RESEARCH

Aerodynamics and Planetary Exploration Group (APEX)

Advisor: Dr. Siegfried Eggel

Interstellar Navigation and Star Identification

Sep 2024 - present

- ❖ Developing star identification techniques for interstellar travel at relativistic speeds by combining traditional algorithms with clustering based on stellar classification.

Celestial Navigation to Supplement DSN Tracking

Jul 2024 - present

- ❖ Collaborating with four colleagues to assess feasibility of using celestial navigation to meet cruise phase uncertainty requirements in heliocentric and cislunar space.
- ❖ Designing sensor tasking/scheduling algorithms with novel considerations for covariance propagation and visibility constraints over the time domain.
- ❖ Implemented spherical harmonics for an open source orbit propagator in C++.

Lost in Space and Time Navigation using Variable Stars

Nov 2022 - present

- ❖ Developed an autonomous optical navigation technique based on light curve analysis of variable stars which does not require prior knowledge of position or time.
- ❖ Simulated star tracker observations and performed Monte Carlo analysis to assess the feasibility of reestablishing Earth pointing via the proposed method.

Aerospace Mission Analysis Laboratory

Advisor: Dr. Zachary R. Putnam

Lost in Space X-ray Pulsar Navigation (XNAV)

Jun 2021 - present

- ❖ Developed algorithms to resolve spacecraft position and time from pulsar timing data.
- ❖ Created a photon time of arrival simulation tool in Python and validated results against the pulsar timing package [PINT](#) to within 50 μ s.
- ❖ Demonstrated the computational feasibility of (XNAV) in a lost in space scenario.

Velocity-based Initial Orbit Determination

Aug 2019 - Apr 2024

- ❖ Developed a novel initial orbit determination (IOD) technique using sequential range-rate measurements to distant stars.
- ❖ Assessed the performance of velocity IOD methods using Monte Carlo simulation.

Space Systems Optimization Laboratory

Advisor: Dr. Koki Ho

Architecture Trade Studies on IRSU Technologies for Human Space Exploration

Sep 2018 - Sep 2019

- ❖ Worked with two graduate students to discuss the development of a mixed-integer linear programming framework for space logistics optimization.
- ❖ Conducted literature review of in-situ resource utilization (ISRU) technologies to quantify input/output rates of technology options for an architecture optimization tool.

SKILLS

Programming Languages	Python, C/C++, MATLAB, Javascript
Software/Prepackaged Tools	L ^A T _E X, Git, GMAT
Operating Systems	Linux, MacOS, Windows
Languages	English, Mandarin Chinese

HONORS AND AWARDS

Yee Fellowship Grainger College of Engineering, University of Illinois Urbana-Champaign	2021, 2024
Aerospace Engineering Alumni Advisory Board Fellowship Department of Aerospace Engineering, University of Illinois Urbana-Champaign	2024
Best Student Paper, 46th AAS GN&C Conference American Astronautical Society (AAS) Rocky Mountain Section	2024
David and Catherine Thompson Space Technology Scholarship American Institute of Aeronautics and Astronautics (AIAA)	2020
H.S. Stillwell Problem Solving Award Department of Aerospace Engineering, University of Illinois Urbana-Champaign	2020
Robert W. McCloy Memorial Award Department of Aerospace Engineering, University of Illinois Urbana-Champaign	2019
James Scholar University of Illinois Urbana-Champaign	2018 - 2021

JOURNAL ARTICLES

L. Hou, I. Bansal, C. Davis, and S. Eggl, “Position and Time Determination without Prior State Knowledge via Onboard Optical Observations of Delta Scuti Variable Stars”, *The Journal of the Astronautical Sciences*, in review (2024).

L. Hou and Z. R. Putnam, “Autonomous Initial Orbit Determination using Sequential Range-Rate Measurements”, *The Journal of the Astronautical Sciences* **71**, 24 (2024).

L. Hou and Z. R. Putnam, “A Norm-Minimization Algorithm for Solving the Lost-in-Space Problem with XNAV”, *The Journal of the Astronautical Sciences* **71**, 6 (2024).

H. Chen, T. Sarton du Jonchay, **L. Hou**, and K. Ho, “Multifidelity Space Mission Planning and Infrastructure Design Framework for Space Resource Logistics”, *Journal of Spacecraft and Rockets* **58**, 538–551 (2021).

H. Chen, T. S. du Jonchay, **L. Hou**, and K. Ho, “Integrated In-Situ Resource Utilization System Design and Logistics for Mars Exploration”, *Acta Astronautica* **170**, 80–92 (2020).

CONFERENCE PROCEEDINGS

L. Hou and S. Eggl, “Autonomous Optical Navigation using Astrometry and Photometry”, in AAS/AIAA Space Flight Mechanics Meeting, AAS 25-248 [accepted] (Jan. 2025).

L. Hou, I. Bansal, C. Davis, and S. Eggl, “Lost-in-space Position and Time Determination via Star Tracker Observations of Periodic Variable Stars (AAS-24-012)”, in *46th Rocky Mountain AAS GN&C Conference* (Feb. 2024).

A. Auster, A. Bojinov, S. Kesarwani, G. Sieck, R. Shah, M. Odeen, S. McCarthy, M. Frankenthor, K. Patel, G. Davis, et al., “Mars Ice Thermal Harvesting Rig & ISRU Laboratory (MITHRIL)”, in *ASCEND 2022* (Aug. 2022), p. 4249.

L. Hou and Z. R. Putnam, “A Norm-minimization Method for Solving the Cold-Start Problem with XNAV”, in AAS/AIAA Astrodynamics Specialist Conference, AAS 22-560 (Aug. 2022).

L. Hou and Z. R. Putnam, “Initial Orbit Determination from Sequential Line-of-Sight Velocity Measurements”, in AAS/AIAA Astrodynamics Specialist Conference, AAS 22-561 (Aug. 2022).

L. Hou, K. Lohan, and Z. R. Putnam, “Comparison and Error Modeling of Velocity-Based Initial Orbit Determination Algorithms”, in AAS/AIAA Space Flight Mechanics Meeting, AAS 21-280 (Feb. 2021).

H. Chen, T. Sarton du Jonchay, **L. Hou**, and K. Ho, “Multi-Fidelity Space Mission Planning and Space Infrastructure Design Framework for Space Resource Logistics”, in *AIAA Propulsion and Energy 2019 Forum* (2019), p. 4134.

H. Chen, T. Sarton du Jonchay, **L. Hou**, and K. Ho, “Integrated Analysis Framework for Space Propellant Logistics: Production, Storage, and Transportation”, *Lunar ISRU 2019-Developing a New Space Economy Through Lunar Resources and Their Utilization* **2152**, 5003 (2019).